Track A: Multi-label Emotion Detection

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Task Description

Objective: Building a **Naive Bayes classifier** to predict emotions (anger, fear, joy, sadness, surprise) from text and evaluate its performance.

Steps:

1. **Exploratory Data Analysis (EDA)**
    - Analyzing emotion label distribution (counts, multi-label frequency).
    - Checking text length statistics.
2. **Preprocessing**
    - Cleaning text (lowercase, remove punctuation, etc.).
    - Tokenizing and vectorizing text (TF-IDF or CountVectorizer).
3. **Modeling**
    - Training a **Multinomial Naive Bayes** classifier for each emotion (binary classification)
    - Adaptation for multi-label prediction if needed.
4. **Evaluation**
    - Assessing performance using metrics (accuracy, precision, recall, F1-score).
    - Comparison of results across emotions (note imbalances).
5. **Conclusion**
    - Summarizing model strengths/weaknesses.
    - Suggesting improvements (handling class imbalance).

Tools:

- Python (Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK).
- Jupyter Notebook for code + brief report.

Motivation: Emotion detection in text can be crucial for many applications like Customer Feedback Analysis, Mental Health Monitoring, Chatbots & Virtual Assistants and so on.

Dataset

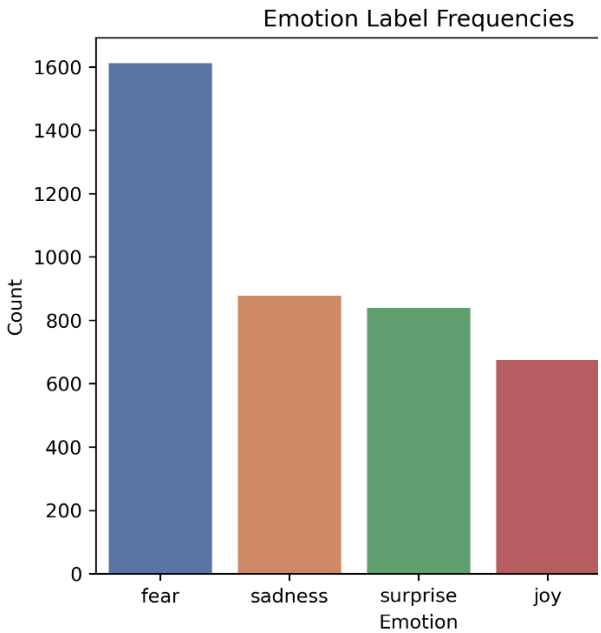
Our explanatory analysis tells us that the given dataset contains text samples labeled with five emotion categories: anger, fear, joy, sadness, and surprise. The dataset consists of 2,768 entries, each representing a text snippet with one or more emotion labels.

1.1 Emotion Distribution

```
Emotion Distribution Analysis
==============================
Total entries: 2768

Absolute counts:                    Percentages:
Anger: 333                          Anger: 12.03%
Fear: 1611                          Fear: 58.20%
Joy: 674                            Joy: 24.35%
Sadness: 878                        Sadness: 31.72%
Surprise: 839                       Surprise: 30.31%
```

We can see this distribution on the barplot as well to visualize results:



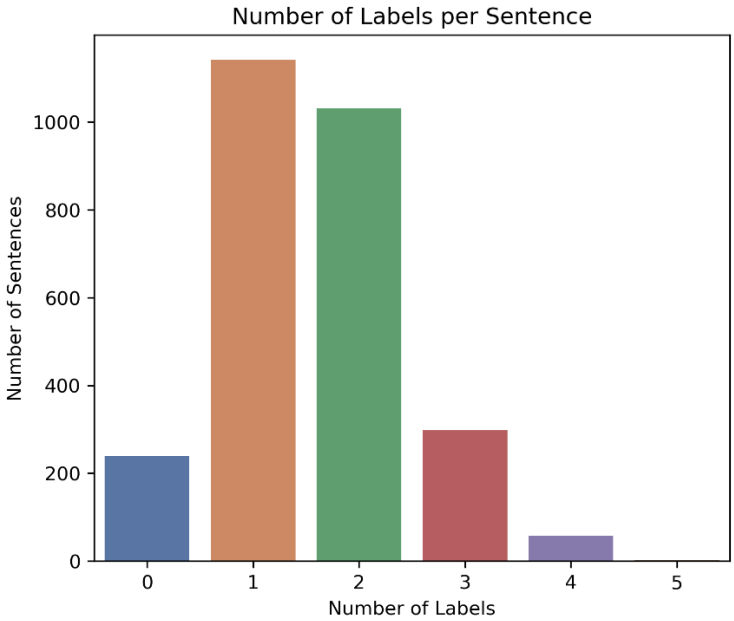
Key Observations

1. Fear is by far the most prevalent emotion in the dataset, appearing in over half of all samples.
2. Anger is the least represented emotion, appearing in only about 12% of samples.
3. The three negative emotions (fear, sadness, anger) collectively appear more frequently than the two positive/neutral ones (joy, surprise).

1.2 Multi-Label Analysis

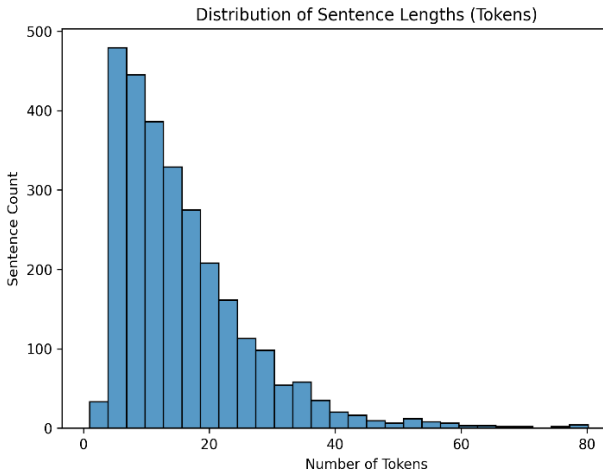
Label Distribution per Sample

1. The majority of samples (55.2%) are labeled with only one emotion.
2. A significant portion (38.5%) have two emotion labels, indicating emotional complexity in many texts.
3. A small percentage (6.3%) exhibit three simultaneous emotions.



1.3 Text Length Characteristics

Token Count Statistics



- count 2768.000000
- mean 15.477601
- std 10.856667
- min 1.000000
- 25% 8.000000
- 50% 13.000000
- 75% 20.000000
- max 89.000000

Key Observations

1. The text samples are relatively short, with a median length of 13 tokens.
2. The distribution is right-skewed, with some substantially longer samples (up to 89 tokens).
3. The standard deviation indicates considerable variability in text lengths.

1.4 Preprocessing

At the end the preprocessing part is started by lowercasing all words and removing punctuations and stopwords from the text samples.

Methodology

1. Data Preparation

- **Input**: Text sentences labeled with emotions (anger, fear, joy, sadness, surprise).
- **Preprocessing**: Text Cleaning**,** Tokenization, Stopword Removal**,** Vectorization

2. Model Design

- **Algorithm**: Multinomial Naive Bayes
- **Approach**:
    - Treating each emotion as a **separate binary classification task** (One-vs-Rest).
    - Training 5 models (one per emotion) due to multi-label possibility.
- **Baseline**: Comparison against a majority class.

3. Training & Evaluation

- **Spliting Data**: 70% train, 30% test.
- **Metrics**:
    - **Per Emotion**: Precision, Recall, F1-score (address class imbalance).
    - **Overall**: Accuracy (using macro & micro averaging).

4. Interpretation

- **Feature Importance**: Extraction of top words (TF-IDF coefficients) per emotion.
    - E.g., "happy" → joy; "scared" → fear.
- **Error Analysis**: Inspecting misclassified examples (e.g., "surprise" vs. "fear").

Evaluation

We should use 2 by 2 **contingency table** for each labeling (e.g., joy, sadness, etc.), construct a separate **confusion matrix**:

	Predicted = 1	**Predicted = 0**
Actual =1	True Positive (TP)	False Negative (FN)
Actual =0	False Positive (FP)	True Negative (TN)

This gives us a per-emotion basis to calculate metrics.

Then we have to calculate Per-Label Metrics. For each label:

- **Precision** = TP / (TP + FP)
- **Recall** = TP / (TP + FN)
- **F1 Score** = 2 * (Precision * Recall) / (Precision + Recall)
- **Accuracy** = (TP + TN) / (TP + TN + FP + FN)

After computing scores for each label, we need to show them into a single performance measure by:

- **Macro Averaging:** Calculating the metric for each label separately, and then take the average
- **Micro Averaging**: Combining all contingency tables: sum TP, FP, FN, and TN across all labels. Then computing the overall precision, recall, and F1 based on these totals